LayerForge v9: Phase 2b Verification Update

Two-Layer Fidelity Structure, Hybrid Ensemble Path, and Pattern Coverage

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Abstract

We report the Phase 2b verification update for LayerForge, a deterministic text decomposition tool that extracts 4±1 hierarchical themes from input text via CPM-based community detection on sentence-transformer embeddings. The v8.1 published version positioned LayerForge as a topic-extraction subtask within an IE pipeline (ADR-026). This v9 update lands 47+ direct verifications across 14 datasets spanning EN/JA, multi-turn dialogue, multi-hop QA, news, and long-form documents. We articulate a two-layer fidelity structure with quintuple PASS evidence at theme-level semantic preservation (response cosine 0.939, BERTScore 0.93, topic coherence UMass-best 6/6 vs LDA/NMF) and quintuple FAIL evidence at fact-level lexical preservation (answer substring 10%, ROUGE-L 0.07, attribute uniform-FAIL, downstream LLM refusal 100%). We confirm a Hybrid ensemble improvement path (LayerForge + K-means) with statistical significance on hotpotqa N=100 (p=2.7e-05) and livedoor N=27 cross-corpus (p=5.1e-08). We complete 14 parameter ablations confirming baseline parameter optimality, and articulate pattern coverage with explicit extrapolation boundaries. We disclose the structural limitation of fact-level fidelity as a constitutive design property (ADR-026 IE pipeline subtask position), not a defect, and motivate the bucket B hybrid pipeline as the operational remedy.

1. Introduction

LayerForge v8.1 was published as a deterministic text decomposition tool that produces 4±1 hierarchical themes from arbitrary input text. The core algorithm combines: (i) sentence-transformer embeddings (paraphrase-multilingual-mpnet-base-v2), (ii) CPM-Louvain community detection on cosine similarity graphs (Traag 2011, Blondel 2008; leidenalg GPLv3 excluded for MIT-clean path), (iii) scale_finder with K=4±1 cognitive constraint (Miller 1956, Cowan 2001), and (iv) ctfidf-based representation extraction (Grootendorst 2022). The intended position is as a topic-extraction subtask within a larger IE pipeline (ADR-026), complemented by NER, regex extraction, and RAG retrieval for fact-level information.

This v9 update reports Phase 2b verification work: 47+ direct verifications + 14 parameter ablations across 14 datasets, formally articulating: (a) what LayerForge can and cannot do, (b) where the structural limitations lie, (c) which improvement paths are evidence-grounded. All verifications follow the pre-register/post-run protocol (ADR-022 §3) with §1-§5 immutability and §6-§8 post-run results. Append-only / immutable archive principles apply throughout.

2. Related Work

Topic modeling baselines: LDA (Blei 2003), NMF (Lee & Seung 1999), BERTopic (Grootendorst 2022), Top2Vec (Angelov 2020). LayerForge differs by enforcing K=4±1 cognitive constraint, requiring no LLM calls (sentence-transformer embeddings only), and being functional on small samples (N=3-10) where BERTopic returns K=0 due to HDBSCAN density failures.

Fidelity proxy metrics: ROUGE-L (Lin 2004), BERTScore (Zhang et al. 2020), topic coherence UMass/NPMI (Mimno 2011, Aletras & Stevenson 2013). We use these as direct fidelity measures rather than reduction-only metrics, articulating a two-layer fidelity structure rather than aggregating.

IE pipeline position: ADR-026 articulates LayerForge as a topic-extraction subtask, complemented by NER (spaCy/GiNZA), regex extraction, and RAG (sentence-BERT retrieval) for entity, numeric, and answer-level information.

3. Method

All verifications follow ADR-022 §3: pre-register §1-§5 (claim, dataset, why-this-data, method, pre-commit interpretations) with immutability, then execute and append §6-§8 (run record, outcome, impact). This prevents post-hoc threshold drift and ensures evidence is honest.

Datasets used: WildChat (EN dialogue), ShareGPT (EN dialogue), LongBench hotpotqa (EN multi-hop QA), loogle (EN long doc), CNN/DailyMail (EN news), livedoor news (JA news), jmultiwoz (JA dialogue), jawiki (JA wiki), Aozora (JA literature), BSD parallel (EN/JA), meetingbank, mt_bench_101, Newman benchmark graphs (polbooks, football).

Subagent fallback: For LLM-as-judge evaluations where formal API key was unavailable, we used Claude Code general-purpose subagent. We acknowledge self-preference bias (Claude evaluating Claude compressed context) as a structural limitation and mark all such evidence as direction-only, requiring formal Anthropic API + Haiku 4.5 re-confirmation for paper-level claims.

4. Results

4.1 Two-Layer Fidelity Structure

Verification across 47+ records reveals a two-layer fidelity structure with internally consistent but oppositely-signed evidence at each layer:

Layer	Metric	Verdict Evidence (V-IDs)	Direction
Theme-level semantic	Response cosine	V-007 (0.939, N=6)	PASS
Theme-level semantic	BERTScore F1	V-025 (0.93 roberta / 0.84 mbert, N=6)	PASS
Theme-level semantic	Topic coherence (UMass)	V-026 (LayerForge best 6/6 vs LDA/NMF)	PASS
Theme-level semantic	Topic coherence (NPMI)	V-026 (+0.044 vs LDA, -0.027 vs NMF, competitive)	PASS
Theme-level semantic	JA title BERTScore	V-029-d (mbert 0.59, delta -0.05 vs baseline)	PASS-RELATIVE
Fact-level lexical	Answer substring	V-021 (10%, N=10, hotpotqa)	FAIL
Fact-level lexical	ROUGE-L vs lead	V-022 (0.074, N=9, livedoor)	FAIL
Fact-level lexical	Attribute breakdown	V-023 (UNIFORM-FAIL across types)	FAIL
Fact-level lexical	Downstream LLM (F3)	V-024 (100% refusal, N=5)	FAIL

Layer	Metric	Verdict Evidence (V-IDs)	Direction
Fact-level lexical	Downstream LLM (F4 hybrid)	V-024-bis (refusal 0/5 + accuracy 0/5, delta -0.60)	FAIL

The two layers are independent dimensions: theme-level semantic preservation is structurally PASS (quintuple-evidence across cosine / BERTScore / topic coherence in EN dialogue, JA news, EN news summary), while fact-level lexical preservation is structurally FAIL (quintuple-evidence across substring, ROUGE-L, attribute, downstream LLM accuracy). This is articulated as a constitutive design property of LayerForge (ADR-026 topic-extraction subtask position), not a defect.

4.2 Hybrid Ensemble Path (V-036 Improvement)

V-036 observed that simple K-means baseline (K=4 + top-10 frequency tokens) achieves tok_recall 2x superior to LayerForge alone on hotpotqa N=5. We pursued Path 2 (LayerForge + K-means ensemble) as the v8.1-integrity-preserving improvement direction. N progression confirms statistical significance:

V-ID	Dataset	N	LF tok	KM tok	Ens tok	Ens vs LF (p)	Ens vs KM (p)
V-042	hotpotqa	5	0.067	0.133	0.133	+0.067 (n.s.)	+0.000 (n.s.)
V-042-bis	hotpotqa	30	0.241	0.297	0.339	+0.098	+0.042
V-042-tri	hotpotqa	100	0.176	0.256	0.290	+0.113 (2.7e-05)	+0.033 (2.4e-03)
V-042-quad	livedoor (JA)	27	0.372	0.252	0.512	+0.140 (5.1e-08)	+0.260 (1.3e-10)

Both datasets confirm ensemble is statistically significantly superior to both baselines (paired t-test, Wilcoxon signed-rank). Cross-corpus direction reversal observed: hotpotqa favors K-means alone (Δ +0.080), livedoor favors LayerForge alone (Δ +0.121, direction reversed), but ensemble is superior in both. This articulates dataset-dependent baseline behavior.

4.3 Parameter Ablation (14 items, all-current-value confirmed)

We exhaustively ablate parameters with the careful strategy (parameter_baseline.md Rule §1-§6): one parameter at a time, baseline snapshot comparison, append-only history, v8.1-integrity-preserving scope where applicable.

Parameter	Ablation	Verdict	Source
tokenizer pattern	Add [0-9]+	NEGATIVE (digit lost in LayerForge internal)	V-027
CPM γ (resolution)	Sweep 0.01-1.0	Dataset-dependent best, default OK	V-031 retro
embedding model	mpnet vs MiniLM-L6	MPNET SUPERIOR (N=30, Δ -0.058)	V-032/ V-032-bis
random_seed	5 seeds variance	ROBUST-STRONG (substr stdev 0.0)	V-033
community method	newman vs cpm	Fact metric invariant	V-034
target_layer_count	(3,5) vs (3,10)	Fact metric invariant	V-035
scale_finder	vs K-means baseline	LF-INFERIOR direction, fixed by V-042 hybrid	V-036
MAX_NODES	25/50/100/200	INVARIANT	V-037

Parameter	Ablation	Verdict	Source
TOP_MEMBERS_PER_LAYER	5 vs 10/20	NO-IMPROVEMENT	V-024-tri
SNIPPET_CHARS	120 vs 240/480	PARTIAL-IMPROVEMENT	V-024-quad
representation_summary	post-process top-K	Cost reduction OK, fact invariant	V-041
token_representations	alone vs union	Redundant overlap with rep_summary	V-041
bridge_nodes	output presence	Output absent in decompose.run, present in prepare_render_data	V-041 (corrected)
Phase 2 thresholds	ADR-019	Driver-level, V-024-tri/quad subset	V-041

All 14 ablations confirm baseline parameter values are at or near the IMPROVEMENT-LARGE threshold boundary (no parameter shows >0.30 delta improvement). Baseline snapshot v1 is preserved unchanged. V-027 finding (NEGATIVE for digit preservation) motivates Path 1 core-spec change (ctfidf → tf hybrid) as v8.1-integrity-trigger future work.

4.4 Pattern Coverage and Extrapolation Boundaries

We map verification coverage across language × preprocessing × length:

Language × Preprocessing	Short (<1k)	Mid (1-10k)	Long (10k+)
EN default	V-002/003/006/007/010/019	V-018/V-029-f/V-012	V-011 + 16 hotpotqa verifications
JA + MeCab	V-015/V-019(BSD)	V-013/V-014/V-022/V-029-b/V-029-d/V-042-quad	V-017/V-019(Aozora)
JA default	V-019 BSD floor -0.05	V-013 floor 0.025	(structural mismatch)
Graph	V-008 polbooks	—	V-009 football
EN formal summary	—	V-029-f CNN/DM	(XSum/SQuALITY future)

Explicit extrapolation boundary: EN long-form is dominated by 16 hotpotqa-derived verifications. Generalization to other EN long-form datasets (LongBench narrativeqa, musique, 2wikimqa) is not directly evidenced; direction is consistent across the hotpotqa-derived measurements but not cross-corpus confirmed. Similarly JA mid is dominated by livedoor; cross-corpus to other JA mid datasets (mlsum_ja, wikipedia-summary-ja) is future work.

5. Discussion

5.1 Cross-corpus Direction Reversal

V-042-tri (hotpotqa) shows K-means tok_recall superior to LayerForge alone (Δ +0.080). V-042-quad (livedoor) shows LayerForge superior to K-means (Δ +0.121, direction reversed). Both datasets confirm ensemble superior to both baselines. This articulates dataset-dependent baseline behavior: K-means frequency tokens excel at multi-hop QA where answer tokens are high-frequency content words; LayerForge theme tokens excel at JA news where coherent thematic decomposition is dominant. The Hybrid ensemble captures both regimes.

5.2 Driver-level vs Core-spec Modifications

Driver-level parameter sweeps (TOP_MEMBERS, SNIPPET_CHARS, MAX_NODES, tokenizer matching) preserve v8.1 integrity. Core-spec modifications (ctfidf → tf hybrid, embedding model swap, K=4±1 relaxation, scale_finder algorithm change) require v8.1 integrity reconsideration and paper update synchronization. We do not undertake core-spec modifications in this update; they are documented in future_plan.md for chai-sovereign trigger.

5.3 Limitations

Statistical: N=100 (V-042-tri) achieves $p<0.005$ but is not population-scale; formal claims require N=1000+ replication. N=5-30 verifications (most ablations) are direction-only.

Self-preference bias: LLM-as-judge evaluations (V-006/V-007/V-024/V-024-bis) used Claude Code subagent (Sonnet/Opus) as evaluator on Claude-compressed context. Formal API + Haiku 4.5 confirmation is pending for paper-level claims.

Coverage bias: EN long-form is hotpotqa-dominated (16/16 verifications), JA mid is livedoor-dominated. Cross-corpus to LongBench narrativeqa/musique and mlsum_ja is unevidenced future work.

Constitutive limitations: Fact-level lexical preservation is structurally FAIL (quintuple-evidence). This is articulated as design property (ADR-026), not defect. The operational remedy is hybrid pipeline with NER/regex/RAG, not LayerForge internal modification.

6. Limitations Summary

- Sample sizes: N=5-100, formal population-scale not achieved
- Self-preference bias: subagent fallback (Claude evaluating Claude) requires formal API confirmation
- Coverage bias: hotpotqa-dominated EN long, livedoor-dominated JA mid
- Fact-level FAIL is constitutive (ADR-026 design), remediated by hybrid pipeline (bucket B)
- Path 1 (core-spec ctfidf → tf hybrid) is unevaluated, requires v8.1 integrity reconsideration

7. Future Work

We articulate 15 future work items across 4 axes (future_plan.md), with explicit chai-sovereign trigger boundary for architectural and scope decisions:

- Axis 1 (Verification, V-101-V-104): Pattern probe accuracy, cross-domain generalization, direction reversal root cause, existing-record reframe
- Axis 2 (Implementation, I-101-I-104): Probe API, routing primitive, output interpretation layer, probe driver isolation
- Axis 3 (Integration, G-101-G-104): Claude Code skill, LangChain/LlamaIndex plugin, LLM API wrapper, KDF integration
- Axis 4 (Real-effective, E-101-E-103): Cost/latency improvement measurement, accuracy preservation, real-world deployment pilot

AI-decidable items: V-101-V-104, I-101/I-103, G-101/G-103, E-101/E-102. chai-sovereign items: I-102 (routing logic = architectural), I-104 (v8.1 integrity trigger), G-104 (KDF integration = two-tool architectural), G-102 (framework expansion = social footprint), E-103 (real deployment).

8. Conclusion

LayerForge v9 confirms a two-layer fidelity structure: theme-level semantic preservation is structurally PASS (quintuple-evidence), fact-level lexical preservation is structurally FAIL (quintuple-evidence). The Hybrid ensemble path (LayerForge + K-means) is statistically significantly superior to both baselines on hotpotqa N=100 ($p<2.7e-05$) and livedoor N=27 ($p<5.1e-08$), confirming the ADR-026 IE pipeline subtask position as the operational remedy. All 14 parameter ablations confirm current baseline parameter optimality; baseline snapshot v1 is preserved unchanged. We articulate explicit extrapolation boundaries (hotpotqa dominance in EN long, livedoor dominance in JA mid) and 15 future work items with chai-sovereign trigger boundaries for architectural and scope decisions.

Acknowledgments

Verification protocol design follows ADR-022 §3 (pre-register/post-run) with §1-§5 immutability principle. Append-only / immutable archive principle follows Microsoft Azure WAF + Fowler + AWS dominant principle for audit-trail-grounded evidence.

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